

SUBHRAJYOTY ROY

 **Nationality:** Indian
 (+1) (314)745-2968
 roy.s@wustl.edu

 [Google Scholar](#)
 github.com/subroy13
 [/in/subroy13](#)
 [Portfolio Website](#)

CURRENT POSITION

- *Postdoctoral researcher* at the Department of Statistics & Data Science in the **Washington University, St. Louis**.

RESEARCH INTERESTS

- Robust statistical inference, divergences, minimum divergence estimation.
- Dependent data analysis, spatial analysis, spatiotemporal analysis.
- Statistical methods for Deep learning, AI safety, and trustworthiness.

EDUCATION

2021 - 2025	Ph.D. in Statistics Thesis: Robust Matrix Factorization using the Density Power Divergence and its Applications. [Link] Supervisors: Ayanendranath Basu and Abhik Ghosh.	Indian Statistical Institute, Kolkata
2019-2021	Masters of Statistics (M.Stat.) Grade: 93.8%, Financial Statistics specialization, Ranked <i>1st</i> .	Indian Statistical Institute, Kolkata
2016 - 2019	Bachelor of Statistics (B.Stat.) Grade: 93.6%, Ranked <i>2nd</i> .	Indian Statistical Institute, Kolkata
2014 - 2016	Higher Secondary (WBCHSE) Grade: 93.4%, District Rank <i>2nd</i> .	Nimta High School, Kolkata
2014	Secondary (WBCSSE) Grade: 94.57%, State Rank <i>20th</i> .	Nimta High School, Kolkata

PROFESSIONAL EXPERIENCE

July, 2025-	Postdoctoral Researcher Department of Statistics and Data Science, Washington University at St. Louis Research Mentor: Prof. Debashis Mondal.	
July, 2022 - June, 2025	Principal Information Researcher Directed applied AI research initiatives, development of RAG-based agentic chatbots (SysCloud Sky & Scout), architecting a centralized control flow system that reduced API rate-limit errors by 90%, etc.	SysCloud Inc.
May, 2024 -July, 2024	Mentor Course instructor for <i>Mathematics for Artificial Intelligence</i> module. Received a consistent rating of 4.5+.	Great Learning Education Series Pvt. Ltd.
July, 2021 - June, 2022	Data Scientist Developed reinforcement learning models to optimize Google Ads bidding strategies, and designed a unified, scalable system architecture for cloud data backup and restoration that drove expansion across major SaaS platforms (Salesforce, Slack, HubSpot, etc.).	SysCloud Inc.

2020 May - June	Data Scientist Intern Sentiment analysis to obtain the impact of COVID-19 on the aviation industry from tweets.	General Electric (Aviation)
2019 May - June	Data Scientist Intern Developed cost optimization techniques for backup and restore workflows. Working on phishing email detection problem using a combination of NLP and deep learning.	SysCloud Inc.

PUBLICATIONS

Journal Articles

1. **Roy, S.**, Sarkar, A., Ghosh, A., & Basu, A. (2026). Asymptotic Breakdown Point Analysis for a General Class of Minimum Divergence Estimators. *Bernoulli*, 32(1), 698-722. [\[Link\]](#).
2. **Roy, S.**, Basu, A., & Ghosh, A. (2024). Robust Principal Component Analysis using Density Power Divergence. *Journal of Machine Learning Research*, 25(324), 1-40. [\[Link\]](#).
3. **Roy, S.**, Ghosh, A., & Basu, A. (2024). Robust singular value decomposition with application to video surveillance background modelling. *Statistics and Computing*, 34(5), 178. [\[Link\]](#).
4. Pyne, A., **Roy, S.**, Ghosh, A., & Basu, A. (2024). Robust and efficient estimation in ordinal response models using the density power divergence. *Statistics*, 58(3), 481-520. [\[Link\]](#).
5. Bhaduri, R., **Roy, S.**, & Pal, S. K. (2022). Rough-Fuzzy CPD: a gradual change point detection algorithm. *Journal of Data, Information and Management*, 4(3), 243-266. [\[Link\]](#).
6. Ghatak, A., Singh Patel, S., Bonnerjee, S., & **Roy, S.** (2022). A generalized epidemiological model with dynamic and asymptomatic population. *Statistical Methods in Medical Research*, 31(11), 2137-2163. [\[Link\]](#).
7. **Roy, S.**, Sengupta, D., Rudra, K., & Saha, U. S. (2020). Analysis of Pollution Patterns in Regions of Kolkata. *Calcutta Statistical Association Bulletin*, 72(2), 133-170. [\[Link\]](#).
8. Mukherjee, D., Dalal, A., & **Roy, S.** (2019). Feasibility of Transparent Price Discovery in Tea through Auction in India. *Commodity Insights Yearbook (MCX)*, 44-52. [\[Link\]](#).

Preprints

1. **Roy, S.**, Basu, S., Ghosh, A., & Basu, A. (2026). Universally Optimal Robustness-Efficiency Tradeoffs for a General Class of Minimum Divergence Estimators. [\[arXiv:2607.04343\]](#). (under review).
2. Kim, B., **Roy, S.**, & Das, P. (2026). BOOOM: Loss-Function-Agnostic Black-Box Optimization over Orthonormal Manifolds for Machine Learning and Statistical Inference. [\[arXiv:2605.04087\]](#). (under review).
3. **Roy, S.**, Deb, S., Karmakar, S., & Roy, R. (2026). Nonparametric regression of spatio-temporal data using infinite-dimensional covariates. [\[arXiv:2604.01593\]](#). (under review).
4. **Roy, S.**, Ghosh, A., & Basu, A. (2025). Robust Rank Estimation for Noisy Matrices. [\[arXiv:2510.19583\]](#). (under review).
5. Bonnerjee, S., **Roy, S.**, & Karmakar, S. (2025). Fast segmentation of watermarked texts from large language models through epidemic change-points framework. [\[arXiv:2509.21160\]](#) (Major Revision from JASA).
6. Jana, S., **Roy, S.**, Ghosh, A., & Basu, A. (2025). Asymptotic breakdown point analysis of the minimum density power divergence estimator under independent non-homogeneous setups. [\[arXiv:2508.12426\]](#). (Major Revision from Test).

7. **Roy, S.**, Basu, S., Ghosh, A., & Basu, A. (2025). Characterization of Generalized Alpha-Beta Divergence and Associated Entropy Measures. [\[arXiv:2507.04637\]](#). (under review).
8. Deb, S., Neves, C., & **Roy, S.** (2024). Nonparametric quantile regression for spatio-temporal processes. [\[arXiv:2405.13783\]](#). (under review).
9. **Roy, S.** (2024). Trustworthy Dimensionality Reduction. [\[arXiv:2405.05868\]](#).
10. Bhaduri, R., Bonnerjee, S., & **Roy, S.** (2019). Onset detection: A new approach to QBH system. [\[arXiv:1908.07409\]](#).

Working Papers

1. **Roy, S.**, Ren, Z. & Yu. M. (2026). Budget-constrained Active Preference Learning for Dueling Bandits with Applications to Prompt Routing.
2. **Roy, S.** & Mondal, D. (2026). Asymptotics of Spatial Graph Laplacians via Sparse Networks under Coulomb's Interaction Law.
3. Mondal, D. & **Roy, S.** (2026). Conditional Formulation of Gaussian Systems on Spatial Networks over Irregularly Distributed Point Sites.

SEMINARS, CONFERENCES & LECTURES

Invited Talks

2026	<i>IISA Conference (upcoming)</i>	Varanasi, India
2026	<i>EcoSta Conference</i>	Kyoto, Japan (virtual)
2026	<i>Statistics & Data Seminar Series</i>	Saint Louis University, USA
2025	<i>Graduate Student Seminar Series</i>	Washington University in St. Louis, USA

Contributed Talks

2026	<i>Poster Presentation (unable to attend)</i>	AISTATS, Marrakesh, Morocco
2025	<i>International Conference in Robust Statistics</i>	Stresa, Italy
2024	<i>Student Paper Competition</i>	IISA Conference, Cochin, India
2024	<i>IMS Asia Pacific Rim Meeting</i>	Melbourne, Australia

Others

Fall, 2022	<i>Guest Lecture on Robust Statistics</i>	Indian Statistical Institute, Kolkata
2021	<i>Prasanta Chandra Mahalanobis memorial lecture</i>	Indian Statistical Institute, Kolkata
2019	<i>D. Basu memorial lecture</i>	Indian Statistical Institute, Kolkata

TEACHING ACTIVITIES

Spring, 26	<i>SDS 5140 Advanced Linear Models</i>	Washington University in St. Louis
Fall, 25	<i>SDS 4010 Introduction to Probability</i>	Washington University in St. Louis
Summer, 24	<i>Mathematics for Artificial Intelligence</i>	Great Learning Education Series Pvt. Ltd.

ADDITIONAL ACTIVITIES

- Maintain a substack newsletter [StatWizard](#) with 200+ subscribers.
- **Session chair:**
 1. IMS Asia-Pacific Rim Meeting, 2024.

• **Reviewer:**

1. Scientific Reports.
2. Applied Stochastic Models in Business and Industry.
3. Colombian Journal of Statistics.

PROJECTS, SOFTWARE AND PACKAGES

2026	ggkodom - Contributor & Maintainer A 'ggplot2'-based toolkit for visualizing individual-level longitudinal trajectories.	CRAN Link
2025	handanim - Contributor & Maintainer A Python library to create whiteboard-style, hand-drawn animations for educational videos, tutorials or data storytelling.	PyPI Link
2024	Langchain - Contributor Framework for developing applications powered by large language models	Github Commits
2023	Multi PDF QueryBot - Developer & Maintainer A streamlit application to chat with PDF documents using RAG and Large language models. Uses a locally hosted Chroma vector database, SQLite for metadata storage, and Claude for access to foundational LLM.	Github Link
2023	callgrid-reader - Developer & Maintainer A typescript utility package to read, parse and analyze callgrind files for profiling analysis of codes	NPM Link
2023	decompy - Developer & Maintainer Python package for multiple robust matrix and tensor decomposition algorithms	PyPI Link
2022	roufcp - Developer & Maintainer Python package implementing a gradual changepoint detection algorithm using rough-fuzzy set theory	PyPI Link
2021	rsvddpd - Developer & Maintainer R package to perform Robust SVD using Density Power Divergence	CRAN Link
2020	COVID-19 Tracker: A Dashboard for prediction and control assistance Developed a dashboard to visualize the COVID-19 situation in different states of India. It dynamically estimates the time-varying reproduction rate and prediction of COVID-19 spread for various lockdown scenarios for district levels in India, based on a modified eSIR model. The project was funded by the Indian Institute of Management, Vishakhapatnam.	
2019	Analysis of Climber's challenges in scaling Mount Rainier Modelling of climber's success rate in climbing Mount Rainier based on weather modelling. Modelling of climber's team sizes for insights about mountaineering-based local economy.	Report Link
2019	An Introduction of Causal Inference using Direct Acyclic Graph Understanding of the setup and assumptions for causal inference. Estimating the effect of intervention using frontdoor and backdoor criterion, simulations and applications to real-world datasets to estimate causality.	Report Link
2018	Chatbot using Seq2Seq Attention Mechanism It is a chatbot created with a seq2seq neural network with a basic attention mechanism, trained with the Cornell Movie Corpus dataset.	Github Link

ACCOLADES

2026	Winner of I2DB Datathon Scientific Symposium	WashU Medicine
2024	Nominated for the Student Paper Award	IISA Conference 2024
2024	AWS Cloud Practitioner Certification	Amazon Web Services (AWS)

2023	AIR 6 at National Eligibility Test Council of Scientific and Industrial Research (CSIR), Govt. of India	
2022	Nominated for P. C. Mahalanobis Memorial Gold Medal Indian Statistical Institute, Kolkata	
2022	J. K. Ghosh Memorial Gold Medal	Indian Statistical Institute, Kolkata
2016-2021	Dean's list of toppers	Indian Statistical Institute, Kolkata
2020	90 percentile at GRE General Test	Educational Testing Service (ETS)
2020	Nikhilesh Bhattacharya Memorial Gold Medal	Indian Statistical Institute, Kolkata
2020	D. Basu Memorial Gold Medal	Indian Statistical Institute, Kolkata
2019	93 percentile at GRE Mathematics Subject Test	Educational Testing Service (ETS)
2018	1st Quartile at Simon Marais Mathematical Competition Australian Mathematical Science Institute	
2016	AIR 1 in Undergraduate Entrance Exam Institute of Mathematics and Application, Bhubaneswar	

SKILLS

Programming Technologies	R, Python, SQL, PHP, JavaScript (NestJS, NodeJS, React, NextJS).
Tools	AWS (S3, Lambda, ECS, Sagemaker, Bedrock), Docker, Langchain, Langflow. MS-Office, $\text{\LaTeX}$

REFERENCES

- **Debashis Mondal.** Associate Professor, Washington University in St. Louis, USA.
Email: mondal@wustl.edu
- **Ayanendranath Basu.** Professor, Indian Statistical Institute, Kolkata.
Email: ayanbasu@isical.ac.in
- **Sayar Karmakar.** Assistant Professor, Department of Statistics, University of Florida.
Email: sayarkarmakar@ufl.edu
- **Diganta Mukherjee.** Professor, Indian Statistical Institute, Kolkata.
Email: diganta@isical.ac.in
- **Vijay Krishna.** CEO, SysCloud Inc.
Email: vijay@syscloudsoftware.com

I certify that the information provided above is true and accurate to the best of my knowledge¹

— Subhrajyoty Roy

¹Last updated on August 13, 2026